

Outliers

① What is outliers

② How to find outliers

③ What is IQR $\rightarrow$ Inter Quantile Range

④ Box Plot for Outliers

⑤ Five Number Summary

* Five Number Summary

① Minimum

② First Quantile (25%) Q_1

③ Median

④ Third Quantile (75%) Q_3

⑤ Maximum

} Box plot

-10
 $D = \{ 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, \underline{27} \}$

* Lower Fence

* Upper fence

$$* \text{Lower Fence} = Q_1 - 1.5(IQR)$$

$$* \text{Upper Fence} = Q_3 + 1.5(IQR)$$

$$* IQR = Q_3 - Q_1$$

$$Q_1 = \frac{25}{100} \times (20)$$

$$= 5 \rightarrow \text{Index}$$

$$\Rightarrow 3$$

$$Q_3 = \frac{75}{100} \times (20)$$

$$\Rightarrow 15 \rightarrow \text{Index}$$

$$\Rightarrow 7$$

$$IQR = Q_3 - Q_1$$

$$= 7 - 3$$

$$\Rightarrow 4$$

$$LF \Rightarrow Q_1 - 1.5(IQR)$$

$$\Rightarrow 3 - 1.5 \times (4)$$

$$\Rightarrow -3$$

$$UF \Rightarrow Q_3 + 1.5(IQR)$$

$$\Rightarrow 7 + 1.5 \times (4)$$

$$\Rightarrow 13$$

① Minimum = 1

② First Quartile (25%) $Q_1 = 3$

③ Median = 5

④ Third Quartile (75%) $Q_3 \Rightarrow 7$

⑤ Maximum = 9

